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Slot-C

QUESTIONS 15 / 20 completed

Decimal to Binary (v)

DECIMAL TO BINARY CONVERSION

Decimal to Binary Conversion: Develop a function to convert a decimal number to its binary equivalent.

PROGRAM EDITOR

C++ v

Run Save

```
1 #include <iostream>
2 using namespace std;
3 void decimalToBinary(int n) {
4     if (n == 0) {
5         cout << "0";
6         return;
7     }
8     int binary[32];
9     int i = 0;
10    while (n > 0) {
11        binary[i] = n % 2;
12        n = n / 2;
13        i++;
14    }
15    cout << "Binary: ";
16    for (int j = i - 1; j >= 0; j--) {
17        cout << binary[j];
18    }
19    cout << endl;
20 }
21
```

OUTPUT

Binary: 10011

INPUT

19

QUESTIONS 15 / 20 completed

Array Sorting (v)

ARRAY SORTING

Array Sorting:
Implement functions to sort arrays

PROGRAM EDITOR

C++ v

Run Save

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     int n;
5     cin >> n;
6     int a[n];
7     for (int i = 0; i < n; i++) cin >> a[i];
8     for (int i = 0; i < n - 1; i++) {
9         for (int j = 0; j < n - i - 1; j++) {
10            if (a[j] > a[j + 1]) {
11                int temp = a[j];
12                a[j] = a[j + 1];
13                a[j + 1] = temp;
14            }
15        }
16    }
17    for (int i = 0; i < n; i++)
18        cout << a[i] << " ";
19    return 0;
20 }
```

OUTPUT

1 4 5 8 9

INPUT

5
8 5 9 1 4

QUESTIONS 15 / 20 completed

String Reversal ▾

STRING REVERSAL

String Reversal:
Design a function that
reverses a given
string.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 string reverseString(string s) {
6     int i = 0, j = (int)s.size() - 1;
7     while (i < j) {
8         swap(s[i], s[j]);
9         i++;
10        j--;
11    }
12    return s;
13 }
14
15 int main() {
16     string s;
17     getline(cin, s);
18     cout << reverseString(s);
19     return 0;
20 }
```

OUTPUT

aynatiahc

INPUT

chaitanya

QUESTIONS 15 / 20 completed

Quadratic Equation ▾

QUADRATIC EQUATION SOLVER

Quadratic Equation
Solver: Create a
function to solve
quadratic equations
($ax^2 + bx + c = 0$)
given coefficients a,
b, and c.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include <iostream>
2 #include <cmath>
3 using namespace std;
4 int main() {
5     double a, b, c;
6     cin >> a >> b >> c;
7     double d = b*b - 4*a*c;
8     if (d > 0) {
9         cout << (-b + sqrt(d)) / (2*a) << " ";
10        cout << (-b - sqrt(d)) / (2*a);
11    }
12    else if (d == 0) {
13        cout << -b / (2*a);
14    }
15    else {
16        cout << "No real roots";
17    }
18    return 0;
19 }
```

OUTPUT

-2 -3

INPUT

1 5 6

QUESTIONS 15 / 20 completed

Temperature Convi ▾

TEMPERATURE CONVERTER

Temperature Converter:
Write functions to
convert temperatures
between Celsius and
Fahrenheit.

Use With Argument and
With Return Value.

PROGRAM EDITOR

C++ ▾

Run

Save

```
1 #include<iostream>
2 using namespace std;
3 float ctof(float c){
4     return c*9/5+32;
5 }
6 float ftoc(float f){
7     return (f-32)*5/9;
8 }
9 int main(){
10     float t;
11     cin >> t;
12     cout << "fahrenheit =" << ctof(t) << endl;
13     cin >> t;
14     cout << "celsius =" << ftoc(t) << endl;
15     return 0;
16 }
```

OUTPUT

```
fahrenheit =77
celsius =37
```

INPUT

```
25
98.6
```